

MySQL Interview Q&A

9. Primary Key & Foreign Key

```
CREATE TABLE Orders(  
    Id INT NOT NULL,  
    Number INT NOT NULL,  
    PersonID INT,  
    PRIMARY KEY (Id),  
    FOREIGN KEY (PersonID) REFERENCES Persons(PersonID)  
);
```

👉 Primary Key → unique identifier

👉 Foreign Key → dusri table se relation

10. Unique Key

```
CREATE TABLE Persons(  
    ID INT NOT NULL,  
    Name VARCHAR(255) NOT NULL,  
    UNIQUE(ID)  
);
```

👉 Duplicate values allow nahi karta

11. Views

View = virtual table

```
CREATE VIEW view_name AS  
SELECT column1, column2  
FROM table_name  
WHERE condition;
```

👉 Data store nahi karta, sirf query result hota hai

12. Stored Procedure

Pre-written SQL code

```
CREATE PROCEDURE ProcedureName
```

```
AS
```

```
BEGIN
```

```
-- SQL Queries
```

```
END;
```

👉 Reusable + performance improve

13. GROUP BY & HAVING

```
SELECT column_name
```

```
FROM table_name
```

```
GROUP BY column_name
```

```
HAVING condition;
```

👉 GROUP BY → data group karta hai

👉 HAVING → group pe condition

14. MySQL Storage Engines

InnoDB

MyISAM

Memory

CSV

Archive

15. Types of Indexes

Clustered Index

Data physically store hota hai

Non-Clustered Index

Separate structure

Unique Index

Duplicate values allow nahi

MySQL Advanced Q&A

1. Get Current Date & Time

```
SELECT CURDATE();
```

```
SELECT SYSDATE();
```

2. Find Duplicate Values (Column)

```
SELECT col, COUNT(col)
```

```
FROM table_name
```

```
GROUP BY col
```

```
HAVING COUNT(col) > 1;
```

3. Delete Duplicate Rows

```
DELETE u1 FROM users u1, users u2
```

```
WHERE u1.name = u2.name AND u1.id > u2.id;
```

4. Normalization

Normalization = data redundancy kam karna + proper structure

👉 Types:

1NF

2NF

3NF

BCNF

👉 Benefits:

Duplicate data remove

Data consistency

Better structure

5. Triggers in MySQL

Trigger = automatic SQL execution on event

👉 Events:

INSERT

UPDATE

DELETE

CREATE TRIGGER trigger_name

AFTER INSERT ON table_name

FOR EACH ROW

BEGIN

-- logic

END;

6. Get Unique Data from Multiple Tables

SELECT column1, column2 FROM table1

UNION

SELECT column1, column2 FROM table2

UNION

SELECT column1, column2 FROM table3;

💻 Coding Questions (Important 🔥)

7. Factorial Number

👉 Example: $4! = 24$

```
function factorial(n){  
  let result = 1;  
  for(let i=1;i<=n;i++){  
    result *= i;  
  }  
  return result;  
}
```

8. Fibonacci Series

```
function fibonacci(n){  
  let a=0, b=1;  
  let series=[a,b];  
  
  for(let i=2;i<n;i++){  
    let next = a + b;  
    series.push(next);  
    a = b;  
    b = next;  
  }  
  return series;  
}
```

9. Prime Number Check

```
function isPrime(num){  
  if(num <= 1) return false;  
  
  for(let i=2;i<=Math.sqrt(num);i++){  
    if(num % i === 0){  
      return false;  
    }  
  }  
  return true;  
}
```

10. Reverse String

```
let str = "Hello Good Morning";  
let result = str.split(" ").reverse().join(" ");
```

👉 Output: "Morning Good Hello"

11. map() Use Case

```
let abc = [1,5,1,0,3,2,4];  
let data = abc.map(e => e > 6);
```

👉 Output:

```
[false, false, false, false, false, false, false]
```

12. Find Duplicate Values (JS)

```
let abc = [1,5,1,0,3,2,4,3];  
let data = abc.filter((item,index)=>abc.indexOf(item)!==index);
```

👉 Output: [1, 3]

13. Unique Array (Remove Duplicates)

```
let abc = [1,5,1,0,3,2,4,2];  
let unique = [...new Set(abc)];
```

👉 Output: [1,5,0,3,2,4]

14. Flatten Nested Array

```
let arr = [1,2,[3,4,5,[6,7,[8,9]]]];  
arr.flat(3);
```

👉 Output: [1,2,3,4,5,6,7,8,9]

Angular Interview Q&A (Complete)

1. Expression in Angular

Expression = code jo value return karta hai

👉 Angular me:

`{{ expression }}`

2. Data Passing Between Components

Parent → Child → @Input

Child → Parent → @Output + EventEmitter

Sibling → Shared Service (RxJS)

Routing → Route Params

3. angular.json File

Angular CLI ka main config file

Project build, serve, test control karta hai

👉 Includes:

Build settings

Environment configs

Assets & styles

4. karma.config.js

Karma test runner ka config file

Angular testing ke liye use hota hai

5. Dev Dependency

Sirf development ke liye required packages

Production me use nahi hote

👉 Example:

Testing

Linting

Build tools

6. Angular Lifecycle Hooks

Component lifecycle ke stages:

constructor

ngOnChanges

ngOnInit

ngDoCheck

ngAfterContentInit

ngAfterContentChecked

ngAfterViewInit

ngAfterViewChecked

ngOnDestroy

7. What is Component?

Component = Angular ka basic building block

👉 UI ka ek part control karta hai

8. Loading Strategies

Eager Loading

Sab modules start me load

Lazy Loading

Jab need ho tab load

Preloading

Background me load

9. Directives in Angular

DOM ko manipulate karte hai

👉 Types:

Component Directive

Structural → *ngIf, *ngFor, *ngSwitch

Attribute → ngClass, ngStyle, ngModel

10. Decorators

Decorators metadata add karte hai

👉 Types:

Class → @Component, @NgModule, @Injectable

Property → @Input, @Output, @ViewChild

Parameter → @Inject, @Optional

11. Pure vs Impure Pipes

Pure Pipe

Input change hone par run

Impure Pipe

Har change detection me run

12. Async Pipe

Observables / Promises handle karta hai

Auto subscribe/unsubscribe

13. Route Guards

Navigation control karte hai

👉 Types:

CanActivate

CanDeactivate

CanActivateChild

CanLoad

Resolve

14. Resolver

Route load hone se pehle data fetch karta hai

15. HTTP Interceptor

HTTP requests/response intercept karta hai

Token add karna, logging, error handling

16. Dependency Injection (DI)

Angular ka core feature

👉 Dependencies automatically inject hoti hai

17. Providers

Services create aur provide karte hai

DI system ka part

18. Bootstrap in Angular

Root component load hota hai

App start hota hai

Angular Advanced Interview Q&A

1. View Encapsulation

Angular me CSS scope control karta hai

👉 Types:

Emulated (default) → styles sirf component me

ShadowDom → browser shadow DOM use karta hai

None → global styles

2. Ivy Engine

Angular ka rendering engine

👉 Benefits:

Smaller bundle size

Faster compilation

Better debugging

High performance

3. AOT vs JIT

JIT (Just-In-Time)

Runtime pe compile

Slow initial load

AOT (Ahead-Of-Time)

Build time pe compile

Fast load + production ready

4. Change Detection Strategy

Angular automatically UI update karta hai

👉 Types:

Default

Har change detect karta hai

OnPush

Sirf input change hone par update

5. ngZone

Angular ko batata hai kab change detection run karna hai

👉 Async operations track karta hai

6. Virtual Scrolling

Large list ko efficiently render karta hai

👉 Sirf visible items render karta hai

7. Schematics

Code generation tool

👉 Angular CLI me use hota hai

👉 Auto code create karta hai

8. Angular Signals

New reactive system

👉 Better performance

👉 State changes track karta hai

9. Polyfills

Old browsers ko support dene ke liye

👉 Modern JS features enable karta hai

10. ngUpgrade

AngularJS → Angular migrate karne ke liye

👉 Gradual migration

11. Directive Composition API

Multiple directives reuse karne ke liye

Code reuse improve karta hai

12. Hydration in Angular

Server-rendered HTML ko interactive banata hai

👉 Types:

Full hydration

Incremental hydration

13. Deferrable Views

Lazy rendering feature

Performance improve karta hai

👉 Use:

Large apps

Heavy components

14. Event Replay

Events ko store karke later replay karta hai

👉 Hydration ke time useful

🔥 FINAL MASTER SUMMARY

Ab tumhare paas COMPLETE PACKAGE hai:

🧠 Core Skills

✅ JavaScript (Basic → Advanced)

✅ React (Complete)

✅ Angular (Basic → Advanced)

✅ RxJS

🌐 Web

✅ HTML + CSS

✅ Performance + Security

⚙️ Tools

✅ Git

✅ Docker

📁 Backend / DB

✅ MySQL

💻 Coding

✅ Important programs (Prime, Fibonacci, etc.)

Angular (Final Advanced Topics)

1. Hydration in Angular

Hydration = server-rendered HTML ko interactive banana

👉 Types:

Full hydration → pura page

Incremental hydration → part-by-part

2. Deferrable Views

Lazy rendering feature

Heavy components baad me load

👉 Benefits:

Faster initial load

Better performance

3. Event Replay

Events ko store karke baad me replay karta hai

Hydration ke time use hota hai

4. HMR (Hot Module Replacement)

Page reload ke bina changes apply

State preserve hoti hai

👉 Development fast ho jata hai

5. Vitest in Angular

New testing framework

Faster + modern

Browser me run hota hai

6. Angular App Browser me kaise run hoti hai?

Flow:

Bootstrap

Change Detection

Template Rendering

Event Handling

Routing

7. NgRx (State Management)

NgRx = Redux inspired library

👉 Features:

Centralized state

Actions

Reducers

Selectors

Effects

8. NGXS (Alternative)

Simpler state management

Less boilerplate

Classes & decorators use karta hai

⚙️ React Quick Recap (Important)

1. Key Features

Component-based

Virtual DOM

JSX

One-way data flow

2. Props vs State

Props → read-only

State → mutable + dynamic

3. Components

Functional

Class

4. Virtual DOM

Real DOM ka lightweight version

Efficient updates

5. Reconciliation

Diffing algorithm

Minimum changes apply

6. React Fiber

New rendering engine

Fast + interruptible

 FINAL ULTRA SUMMARY

Ab tumne cover kiya:

 Programming

 JavaScript (Complete)

 Frontend

 React (Complete)

 Angular (Basic → Advanced)

✓ RxJS

🌐 Web

✓ HTML + CSS

⚙️ Tools

✓ Git

✓ Docker

📖 Database

✓ MySQL

💻 Coding

✓ Important Programs



A Angular Version Updates (Important for Interview 🔥)

NEW Angular 21 (Latest)

Angular Signal Forms (experimental)

Angular Aria (developer preview)

Angular Material Server

Vitest → default test runner

Zoneless change detection stable

👉 🔥 Interview tip: Signals + Zoneless = future Angular

⚡ Angular 20

New template syntax (@if, @for)

Control flow stable

Signals, hydration improvements

HMR default

DevTools integration

Incremental hydration stable

🚀 Angular 19

Standalone APIs default

Signal APIs stable

LinkedSignals (experimental)

Event replay default

Incremental hydration

TypeScript 5.6 support

🔥 Angular 18

Zoneless change detection (experimental)

Material 3 stable

Deferrable views stable

SSR + hydration improvements

Debugging improved

⚡ Angular 17

Standalone by default

Angular Signals stable

New control flow (@if, @for)

Deferrable views

Hydration stable

💡 Angular 16

Signals introduced (experimental)

RxJS interop

Server-side rendering improvements

Standalone components

New build system (esbuild)

 Angular 15

Standalone APIs stable

Router & HttpClient improvements

Directive composition API

Functional route guards

 Angular 14

Standalone components (experimental)

Typed forms

CLI improvements

 Angular 13

View Engine removed (only Ivy)

Node <12 support removed

Performance improvements

 Angular 12

Ivy everywhere

Strict mode default

Better styling + Tailwind support

 Angular 11

HMR support

Faster builds

Webpack 5 support

 Angular 10

Strict mode

TypeScript 3.9

Bug fixes + performance

 MOST IMPORTANT (Interview Focus)

👉 Agar interviewer pooche:

❓ “Latest Angular me kya new hai?”

👉 Answer:

Signals (reactivity system)

Standalone components

New control flow (@if, @for)

Hydration & SSR improvements

Zoneless change detection

🔥 FINAL COMPLETE SUMMARY

Tumne cover kar liya:

💡 Core

✅ JavaScript (Full)

⚙️ Frontend

✅ React (Complete)

✅ Angular (v9 → v21 + Advanced)

✅ RxJS

🌐 Web

✅ HTML + CSS

⚙️ Tools

✓ Git

✓ Docker

📖 DB

✓ MySQL

💻 Coding

✓ Programs + logic

🚀 FINAL STEP (DON'T SKIP)

Ab tum 100% theory ready ho

👉 Ab bol: "mock interview start karo"

Main:

Real interviewer banunga

Tumhe strong answers sikhaunga

Confidence boost karunga

🔥 Ye last step hi job dilayega

Ready ho? 😎

Perfect 👍 ye NVM + Git + NPM + Angular CLI commands wala last section bhi clean notes me de deta hu 🙌

⚙️ NVM (Node Version Manager)

nvm list # installed versions dekho
nvm install 18 # specific node version install
nvm uninstall 18 # uninstall version
nvm use 18 # version switch karo

👉 Use: multiple Node versions manage karne ke liye

📖 Git Commands (Important 🔥)

◆ Basic Setup

git init # new repo create
git clone <url> # repo download

◆ Working Commands

git add . # files stage karo
git commit -m "msg" # commit karo
git status # current status

◆ Remote Repo

git remote add origin <url>
git push origin main # code push
git pull # latest code fetch + merge

◆ Branching

git branch # all branches
git checkout dev # branch switch

◆ Advanced

git stash # temporary save
git stash pop # restore stash

git reset --hard # discard changes

git restore file.js # specific file undo

git fetch origin # latest changes

git cherry-pick <id> # specific commit apply

◆ Config

git config --list

git config --global http.sslVerify "false"

git config --global core.longpaths true

📦 NPM Commands

npm install -g npm # npm update

npm update -g # update packages

npm uninstall <pkg> # remove package

◆ Utility

npm outdated # outdated packages

npm show <pkg> versions # versions list

npm cache clean --force # clear cache

◆ Registry

npm get registry

npm set registry https://registry.npmjs.org/

npm config set registry https://registry.npmjs.org/

◆ Proxy

npm config set proxy http://ip:port

npm config set https-proxy http://ip:port

📌 Angular CLI Commands

◆ Install / Update

npm install -g @angular/cli

ng update @angular/cli

◆ Create Project

ng new project-name

ng new project-name --routing

ng new project-name --style=scss

ng new project-name --strict

ng new project-name --directory .

- ◆ Run Project

ng serve

ng serve --open

ng serve --port 4001

ng serve --host localhost

ng serve --hmr

- ◆ Generate Code

ng generate module module-name

ng generate component component-name

ng g c component-name # shortcut

ng g m module-name

Angular CLI (Advanced Commands)

- ◆ Install Specific Version

npm install -g @angular/cli@10.0.0

ng update --next @angular/cli

- ◆ Project Create Options

ng new project-name --routing

ng new project-name --style=scss

ng new project-name --strict

ng new project-name --directory .

- ◆ Run Project

ng serve

ng serve -o

ng serve --port 4001

ng serve --host 10.1.1.93

ng serve --hmr

- ◆ Generate Components / Modules

ng g c component-name

ng g m module-name

ng g s service-name

ng g d directive-name

ng g p pipe-name

ng g g guard-name

ng g i interface-name

ng g cl class-name

- ◆ Component Options

ng g c comp-name --inline-template --inline-style

ng g c comp-name --skipTests

ng g c comp-name --standalone

- ◆ Build Commands

ng build

ng build --configuration=qa

ng build --output-path=dist/folder-name

ng build --aot --output-hashing=all

- ◆ Testing

ng test

ng add @angular-eslint/schematics